VAVR vs Zazr Deep Dive

eason: 7 · Advanced FP · Category: Functional · Level: Advanced · Build with Waq

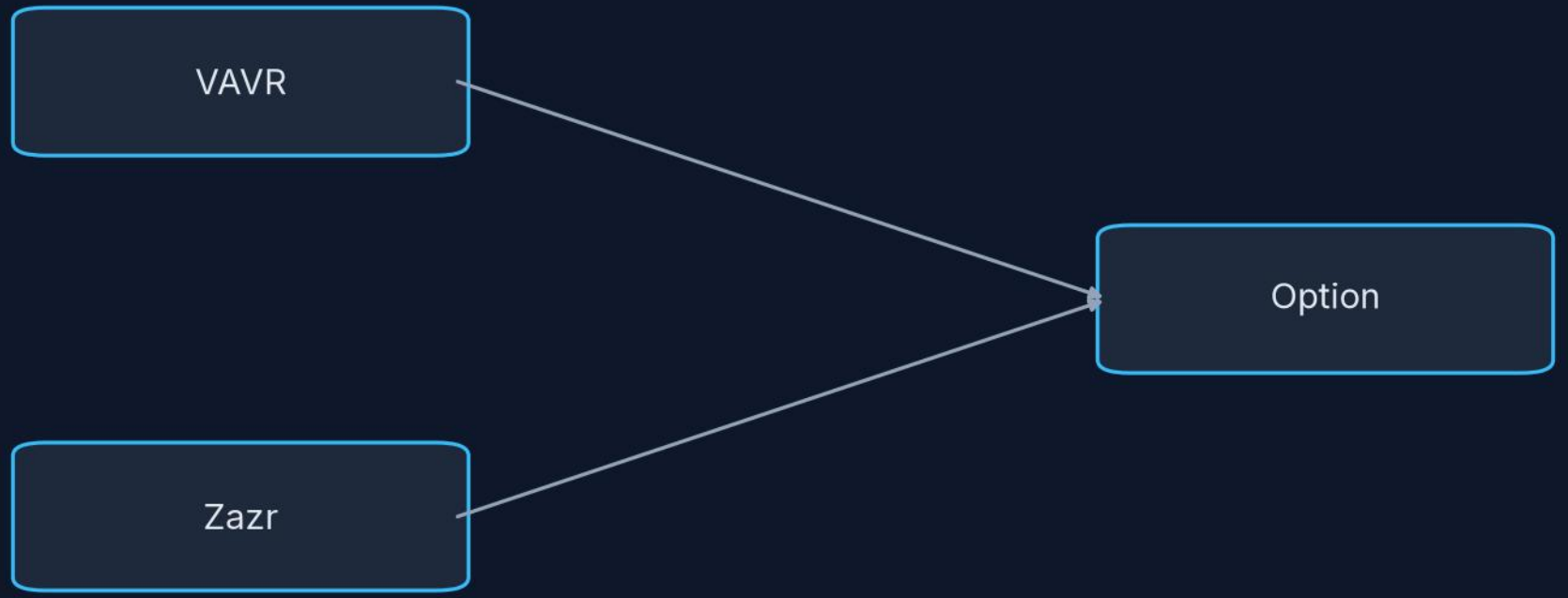
Today's Hook

- Today: VAVR vs Zazr Deep Dive as a StreamForge product force — not trivia.
- Runnable Demo on JDK 25 with captured stdout.
- Classic encoding → Java 25 / Zazr modernization → trade-offs.
- CapCut slides + full spoken script included.

The Problem

- Naive local fixes: more ifs, statics, nulls, or threads.
- Illegal states still compile; tests become order-dependent.
- Design reviews lack a shared name for the force.
- Interviews punish memorization without judgment.

Structure



VAVR vs Zazr Deep Dive structure

Real-World Scenario

- StreamForge creator platform is the running metaphor.
- Same names appear in naive code, refactor, and Demo output.
- Stay in one story for the full 10–12 minutes.
- Product language first; pattern vocabulary second.

Naive Code

```
// Naive sketch for VAVR vs Zazr Deep Dive
public class Naive {
  // duplicated logic / leaked details / unsafe sharing
  Object doIt(Object... args) {
    if (args == null) return null;
    return args[0];
  }
}
```

Why Naive Hurts

- Coupling spreads across controllers and infrastructure.
- Tests need sleeps or static reset hooks.
- New variants require shotgun edits.
- On-call cannot reason about failure modes.

Implementation

```java
// file: src/main/java/com/buildwithwaqar/patterns/functional/vavrcompare/VavrVsZazr.java
package com.buildwithwaqar.patterns.functional.vavrcompare;

/** Side-by-side Option usage — VAVR (classic) vs Zazr (Java 25 sealed/records). */
public final class VavrVsZazr {
    private VavrVsZazr() {}

    public static String vavrDescribe(io.vavr.control.Option<Integer> opt) {
        return opt.map(n -> "got " + n).getOrElse("missing");
    }

    public static String zazrDescribe(dev.zazr.control.Option<Integer> opt) {
        return switch (opt) {
            case dev.zazr.control.Option.Some(var n) -> "got " + n;
            case dev.zazr.control.Option.None() -> "missing";
        };
    }
}
```

Demo.main

```
package com.buildwithwaqar.patterns.functional.vavrcompare;
public final class Demo {
  public static void main(String[] a) {
    System.out.println("=== VAVR vs Zazr walkthrough ===");
    System.out.println("vavrSome=" + VavrVsZazr.vavrDescribe(io.vavr.control.Option.some(1)));
    System.out.println("vavrNone=" + VavrVsZazr.vavrDescribe(io.vavr.control.Option.none()));
    System.out.println("zazrSome=" + VavrVsZazr.zazrDescribe(dev.zazr.control.Option.some(1)));
    System.out.println("zazrNone=" + VavrVsZazr.zazrDescribe(dev.zazr.control.Option.none()));
  }
}
```

Walkthrough Output

```
=== VAVR vs Zazr walkthrough ===
vavrSome=got 1
vavrNone=missing
zazrSome=got 1
zazrNone=missing
```

Step-by-Step Trace

- Each stdout line maps to a refactor stage.
- Re-run Demo locally on JDK 25 to verify.
- Differences usually mean a dirty tree — pull main.
- Pause the video and predict the next line before revealing.

Modern Java 25 / Zazr

```
// Java 25 / Zazr direction for VAVR vs Zazr Deep Dive
// records + sealed + switch
// Option/Either/Validation instead of null
// virtual threads for blocking fan-out
// inject interfaces at the edge
```

Trade-offs

- Clarity vs ceremony — do not spawn five types for two lines.
- Closed sealed worlds vs open SPI plugins.
- Teaching demos omit production idempotency/metrics.
- Prefer simpler language features when they erase the force.

Related Patterns

- Contrast with the neighbor pattern (force difference).
- Name the roles from memory in sixty seconds.
- Point to the season map for what comes next.
- Link the GitHub episode folder in the description.

Recap

- Force → naive pain → VAVR vs Zazr Deep Dive → Demo stdout.
- Modernize with Java 25 / Zazr when it shrinks code.
- Teach when NOT to use it.
- Practice: break a test, fix it, re-run Demo.